EEE301 Signals and Systems
Laboratory Practical 5
Analysis of LTI Systems in *s*-Domain

Title: Laplace transform with MATLAB/OCTAVE

Instruction:
Use MATLAB/Octave to complete the following laboratory exercises.

Theory

The inverse Laplace transform $F(s)$ of a rational function polynomials can be found mathematically by first expanding the function into its partial fractions and then carried out the inverse Fourier transform on individual partial fractions. See the example below.

Example

Find the inverse Laplace transform of a function

$$F(s) = \frac{3s+2}{s^2+3s+2}$$

Solution:

$$F(s) = \frac{3s+2}{s^2+3s+2} = \frac{3s+2}{(s+1)(s+2)}$$

$$\frac{3s+2}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

Multiply left and right sides of the equation with $(s+1)(s+2)$ to obtain

$$3s + 2 = A(s+2) + B(s+1)$$

When $s = -1$,
$$3(-1) + 2 = A(-1+2) + B(-1+1)$$
$$-1 = A + 0$$
$$A = -1$$

When $s = -2$,
$$3(-2) + 2 = A(-2+2) + B(-2+1)$$
$$-4 = 0 + B(-1)$$
$$B = 4$$

Therefore,

$$F(s) = \frac{3s+2}{s^2+3s+2} = \frac{3s+2}{(s+1)(s+2)} = \frac{-1}{s+1} + \frac{4}{s+2}$$

Perform Laplace transform on individual partial fractions:

$$\frac{-1}{s+1} \leftrightarrow -e^{-t}u(t)$$

$$\frac{4}{s+2} \leftrightarrow 4e^{-2t}u(t)$$

Refer to Laplace transform tables in Appendices for the inverse Laplace transform.

Therefore, the inverse Laplace transform of $F(s)$ is

$$f(t) = (-e^{-t} + 4e^{-2t})u(t)$$

Using MATLAB/OCTAVE to obtain partial fractions

For a partial fraction expansion,

$$\frac{N(s)}{D(s)} = k_1 + k_2 s + \cdots + \frac{r_1}{(s-p_1)} + \frac{r_2}{(s-p_2)} + \cdots + \frac{r_i}{(s-p_i)}$$

We can simplify our calculation steps by applying the MATLAB function

```
[r,p,k] = residue(N,D)
```

where *r* = residue, *p* = poles, *k* = constants at the numerators of partial fractions to get the partial fractions without hand calculation.

For example, in the previous function, *F*(*s*)

$$F(s) = \frac{3s+2}{s^2+3s+2}$$

We first declare the numerator polynomial *N* = [3 2] and the denominator polynomial *D* = [1 3 2], which are the coefficients of the polynomials at the numerator and the denominator of function *F*(*s*), respectively.

Then, we call the function `residue(N, D)` to find the residues *r*, poles *p,* and constants *k*.

```
>> N = [3 2];
>> D = [1 3 2];
>> [r,p,k] = residue(N,D)
```

The function shall return the following results:

```
r =
     4
    -1
p =
    -2
    -1
k =
    []
```

Therefore, we can write the result of partial fraction expansion as,

$$F(s) = \frac{3s+2}{s^2+3s+2} = \frac{r_1}{s-p_1} + \frac{r_2}{s-p_2} = \frac{4}{s+2} + \frac{-1}{s+1}$$

MATLAB/OCTAVE function `roots()`

We can obtain the roots of the denominator using the MATLAB function `roots()`. For example,

```
>> roots([1 3 2]
ans =
     -2
     -1
```

The result above means $(s^2 + 3s + 2) = (s+2)(s+1)$
If you are given the roots of a polynomial, and you want to find the polynomials, use the MATLAB function `poly()`.

For example, given roots $s_1 = -2$ and $s_2 = -1$, find the polynomial:

```
>> poly([-2, -1])
ans =
     [1    3     2]
```

The result above means $(s+2)(s+1) = (s^2 + 3s + 2)$.

MATLAB function `poly()`
In a situation where the Laplace transform's denominator is in a factorized form as follows:

$$F(s) = \frac{3s + 2}{(s + 2)(s + 1)}$$

First, convert the polynomial $(s+2)(s+1)$ into $s^2 + 3s + 2$ using the MATLAB function `poly()`:

```
D = poly([-2, -1]);
```

You should get the result `D = [1 3 2]` which are the coefficients of $s^2 + 3s + 2$.

Then, use the MATLAB function residue to get the partial fractions:

```
[r,p,k] = residue([3, 2],D);
```

This shall yield the partial fractions:

$$F(s) = \frac{3s + 2}{(s + 2)(s + 1)} = \frac{3s + 2}{s^2 + 3s + 2} = \frac{4}{s + 2} + \frac{-1}{s + 1}$$

Exercise

Find the inverse Laplace transform of the following functions by first performing the partial fraction expansion using MATLAB functions `residue()` and `poly()` and then apply the Laplace transform tables in Appendix 1 and 2. Finally, show your MATLAB code and calculations.

1. $F(s) = \frac{3s^2+2s+5}{s^3+12s^2+44s+48}$

```
% Insert MATLAB script here..
```

Perform inverse Laplace transform here:

2. $F(s) = \frac{s+3}{s^3+5s^2+12s+8}$

% Insert MATLAB script here..

Perform inverse Laplace transform here:

3. $F(s) = \frac{s+3}{(s+2)(s+1)^2}$

% Insert script here..

Perform inverse Laplace transform here:

4. $F(s) = \frac{s^2+3s+1}{(s+1)^3(s+2)^2}$

% Insert script here..

Perform inverse Laplace transform here:

5. $F(s) = \frac{s^2+2s+2}{s+1}$

% Insert script here..

Perform inverse Laplace transform here:

Appendix 1
Basic Laplace Transforms

Signal	Transform
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$\delta(t-\tau),\ \ \tau \geq 0$	$e^{-s\tau}$
$e^{-at}u(t)$	$\frac{1}{s+a}$
$t^n e^{at}u(t)$	$\frac{n!}{(s-a)^{n+1}}$
$e^{-at}\cos(bt)\,u(t)$	$\frac{s+a}{(s+a)^2+b^2}$
$e^{-at}\sin(bt)\,u(t)$	$\frac{b}{(s+a)^2+b^2}$

Appendix 2
Laplace Transform Properties

Signal	Bilateral Transform
$ax(t)+by(t)$	$aX(s)+bY(s)$
$x(t-\tau)$	$e^{-s\tau}X(s)$
$e^{s_o t}x(t)$	$X(s-s_o)$
$x(at)$	$\frac{1}{\lvert a\rvert}X(\frac{s}{a})$
$x(t)*y(t)$	$X(s)Y(s)$
$x(t)\,y(t)$	$\frac{1}{j2\pi}X(s)Y(s)$
$-tx(t)$	$\frac{d}{ds}X(s)$
$\frac{d}{dt}x(t)$	$sX(s)-x(0^-)$
$\frac{d^2}{dt^2}x(t)$	$s^2X(s)-sx(0^-)-\frac{dx(0^-)}{dt}$
$\int_{-\infty}^{t}x(\tau)d\tau$	$\frac{X(s)}{s}$